Table 1: Image sources: one row per public release, with its modality and device, its role, the unit the split keeps disjoint, what was scored, its truth and what this work releases from it.

Collection[j]	Modality; device	Role in this work	Distinct subjects (split unit)	Frames, films or renders (train / val / test)	Truth source	Released from it[k]
CT with labels, rendered						
CTSpinoPelvic1K	abdominopelvic CT	level bank[a]	801 patients of 802 records[b] (TCIA patient identifier)	801 renders	release labels: levels, ribs, pelvis	corpus; level weights
TotalSegmentator v2 and v3	whole-body CT	level bank; styled corpus (142 subjects, 17 validation)	1,935 scans[c] (CT scan)	1,170 (v2) and 1,735 (v3) renders	labels mapped by name	corpus; level weights
VerSe 19/20	spine CT	level bank; no accuracy quoted[c]	355 subjects, 374 series (subject)	387 renders; 374 soft-tissue renders	VerSe masks C1–L6; published transitional table	corpus; level weights
Radiographs						
CSXA	cervical lateral	level bank; validation	4,963 individuals (individual)	4,963 films	per-vertebra levels, released	corpus; level weights
BUU-LSPINE	lumbar AP and lateral	level bank; validation	3,600 cases[d] (case number)	3,603 lateral films; the 3,600 AP films in no bank	end-plate lines L1–S1; levels by position	corpus; level weights
NHANES II	lateral survey film	level bank; validation	364 persons among the annotated (sample person)	543 annotated films (542 on disk) of 17,075 release entries[e]	landmarks, released	corpus; level weights
VinDr-SpineXR	AP and lateral	intraoperative fold banks only, training, pseudo-labelled[f]	5,000 studies[c] (study)	2,071 films as 2,632 crops, of 10,466 released images	lesion boxes released; levels pseudo-labelled	corpus (all 10,466 images)
FluoroCap	literature fluoroscopy panels	intraoperative fold banks only, training[f]	not established[g] (source article or video)	950 panels as 2,100 crops, of 29,484 harvested	none released	none: dropped
Real exposures with three-dimensional truth						
C-ArmLSpine	mobile C-arm, Surgio-B51; per-case CBCT	naming cross-validation; held-out detector test	not established; 149 released cases, 148 scored[h] (case = acquisition session)	296 scored: 210 / 30 / 56 (105 / 15 / 28 cases); released 299 spot exposures, two cone-beam sweeps[h]	CBCT, two automatic labellers, agreed levels; pose by registration	corpus; level weights (fold models); no detector weight
DeepFluoro	mobile C-arm, Cios Fusion; cadaver CT	detector training; device parameters	6 specimens (specimen; 4 / 1 / 1)	366 frames: 238 / 24 / 104	CT and C-arm pose per frame	corpus; parameters (40 of 366 frames, R^2 0.905 vs 0.899); detector weights
COBRA2026, Elekta XVI	radiotherapy kV imager; planning CT	detector training; device parameters	145 patients of 467 public Elekta cases[i] (case = patient; 110 / 8 / 27)	3,480 frames: 2,640 / 192 / 648	planning CT, deformably registered; RTK geometry	corpus (692 cases, fetching); parameters (40 of 335, R^2 0.974 vs 0.964); detector weights
Geneva hip	mobile C-arm, BV Pulsera; phantom CT	unseen device, test only; device parameters	1 phantom (none: one object)	116 frames, all test	CT and gold-standard pose	corpus; parameters (40 of 116, R^2 0.952 vs 0.917)
Ljubljana 2D-3D-GS	turntable flat panel, PIXIUM 4600; cadaver CT	device parameters only	1 specimen (not split)	18 frames	CT and gold-standard pose	parameters (14 of 18, R^2 0.679 vs 0.650); no frame

AP = anteroposterior; CBCT = cone-beam CT; CDC = Centers for Disease Control and Prevention; RTK = reconstruction toolkit; TCIA = The Cancer Imaging Archive; level bank = the training set of the level encoder and head; R^2 = coefficient of determination. [a]Physics-styled re-renders of the three CT collections (7,045 over 3,029 subjects; 1,482 over 81 transitional subjects) repeat subjects already counted; bank subject counts are not additive. [b]Record 0533 was a truncated download and is in no bank; its rebuild is a gate G0 repair. [c]No patient identifier in the release: the scan, series or study is the finest recoverable unit; the TotalSegmentator count folds 24 renumbered pairs and three duplicate groups; VerSe series were keyed as subjects in the film bank, so no VerSe accuracy is quoted. [d]At most 3,598 by content: two case pairs share byte-identical films. [e]The CDC index lists 17,075 films; two are served empty at the source, so 17,073 are on disk; one annotated film survives only inside the annotation archive. [f]Fold banks = the capped per-fold crop banks of the intraoperative cross-validation (about 31,400 crops per fold); the film bank behind the plain-film accuracies holds no VinDr or FluoroCap image. The released corpus takes all 10,466 VinDr images and drops FluoroCap (a republished figure is the same patient). [g]A panel keys to its source article or video, which may hold several patients or one patient republished. [h]The archive ships 299 spot exposures over 149 cases (149 AP, 149 lateral with one repeat, one oblique) and two 301-projection cone-beam sweeps; the scored bank holds one AP and one lateral of 148 cases (case134's two were missed by a folder walker) and scores case118's −58° oblique as a lateral; rebuilt at gate G0. No patient tag survives; the cases fall on five acquisition dates. [i]692 public cases: 467 Elekta XVI (centres A–D), 225 Varian OBI (F–G); 152 Elekta cases were fetched before the analysis and 145 kept — two pilot cases (A004, A005) never entered the export list and five (A081, A239, A260, A303, B000) failed an axis-map check, B000 because its planning CT is released outside the imaged volume; 26 Varian cases trained no model; the full release is fetching (gate G0). [j]Totals over the additive rows: 4,093 renders (801 + 1,170 + 1,735 + 387); 9,108 level-bank films (4,963 + 3,603 + 542; 2,071 VinDr films and 950 FluoroCap panels sit in the fold banks alone); 4,276 real frames scored (296 + 366 + 3,480 + 116 + 18), of which the mobile C-arm rows (296, 366, 116) give 778; subjects are not totalled because the units differ. [k]Corpus: rebuilt from the public release by the released exporters (the licences differ, so derived frames are not redistributed). Parameters: a fitted nine-term image-formation file for the device, with the frames it was fitted on and its median held-out R^2 over six folds against uncalibrated physics. Weights: the released weights were trained on it (level weights = the fold models of the intraoperative cross-validation; detector = the held-out-device detector trained on COBRA2026 and DeepFluoro real frames); corpus and weights ship at gate G7. Absent above, used by no trained component: the SpinalAI, AASCE and anchor-crop render banks, the SpineSAC phantom frames, Mendeley-716.

Table 2: Transitional persons by subtype: definition, persons per source, the truth behind the subtype, the test side of the built person-disjoint split, the share of held-out persons the current level encoder trained on, the unread COBRA2026 screen calls, and the best-case Wilson 95 % lower bound the test count allows.

Subtype[a]	Definition; what a detector can see[b]	Persons: total (CTSP / VerSe / TS / C-Arm)	Truth behind the subtype	Test persons, built split[c]: CT + real	Validation and test persons the current encoder trained on[d]	Unread COBRA2026 screen calls[e]	Wilson lower bound with no error at the test count, %[f]
L6	six rib-free vertebrae above the sacrum; six lumbar bodies where the sixth carries a label	110 (18 / 86 / 3 / 3)	CTSP two-reader consensus or release label; VerSe published read (38 persons on the audit read alone); TS independent rib count; C-Arm one automatic labeller	43 + 3	49 of 54	38	91.8 (43 CT); 43.9 (3 real)
T13	thirteen rib-bearing vertebrae	7 (0 / 6 / 1 / 0)	VerSe mask label 28; TS independent rib count	3 + 0	4 of 4	45	43.9 (3 CT)
Sacralised L5, four lumbar	L5 fused to the sacrum (Castellvi III or IV) and four lumbar labels; the fused body sits in the sacrum label: one box	12 (9 / 3 / 0 / 0)	CTSP two-reader consensus; VerSe published read	5 + 0	14 of 14, both sacralised rows	8	56.6 (5 CT)
Sacralised L5, five lumbar	L5 fused to the sacrum on a five-lumbar count; its own label (CTSP) or none (VerSe)	16 (3 / 13 / 0 / 0)	as the row above	6 + 0	with the row above	none called	61.0 (6 CT)
Castellvi I/II	enlarged or articulating transverse process; the count is unchanged	31 (4 / 27 / 0 / 0)	CTSP two-reader consensus; VerSe published read	12 + 0	15 of 15	not screened	75.8 (12 CT)
Lumbar rib	rib on L1; the count is unchanged	14 (14 / 0 / 0 / 0)	CTSP release labels	6 + 0	7 of 7	not screened	61.0 (6 CT)
Unassigned	four lumbar labels, source read normal; no subtype established	4 (4 / 0 / 0 / 0)	CTSP release labels	2 + 0	3 of 3	–	–
L6 and T13 in one spine	none in any truth table	0	–	–	–	–	–
Any subtype	the rows above	194 (52 / 135 / 4 / 3)	–	77 + 3	92 of 97	91	–

CTSP = CTSpinoPelvic1K; TS = TotalSegmentator; C-Arm = C-ArmLSpine; L5, L6 = fifth, sixth lumbar vertebra; T13 = thirteenth thoracic vertebra. [a]Distinct persons (patient key; two VerSe series pairs are one person each); rib convention: L6 = six rib-free vertebrae between the last rib-bearing vertebra and the sacrum, T13 = thirteen rib-bearing vertebrae. Counted 23 September 2026 from the subtype table; eight of twelve TotalSegmentator spines once called L6 from raw mask identifiers read normal on an independent rib count and are not counted. [b]Whether a detector sees one body or two is decided by the mask: a fused body with its own label gives two boxes, one inside the sacrum label gives one, one left unlabelled gives none. [c]Person-disjoint, subtype-stratified split, 50 / 10 / 40 train / validation / test within each subtype, every real transitional subject on the test side (the three C-ArmLSpine L6); built 23 September 2026, leak audit passed. The CT test counts equal round(0.40 × persons), the arithmetic Figure **??** draws. [d]Persons on the validation or test side of the built split who sit in the training split of the level bank behind the current encoder (v8 or v9); a model descending from those banks cannot score them, so the transitional comparison needs an arm trained without them. [e]SPINEPS with VERIDAH on the planning CT deformably registered to the cone-beam CT, read by no person; on CTSP this labeller's L6 call matched the readers in 10 of 24. The screen also flagged three cases for re-reading. All calls sit on the test side; none counts as a transitional test person until verified. [f]Wilson 95 % lower bound when every test person is named correctly, the best case at that count; the pre-stated targets are 90 % for the exact level (35 persons with no error) and 85 % for variant sensitivity (22 carriers without a miss).

Table 3: The LevelCheck comparator, frozen before any held-out transitional person was scored: per hypothesis the stratum, the test, the modelled LevelCheck accuracy and its source, the published range, the smallest test count at which an error-free result passes, and the result owed.

Hypothesis[a]	Stratum[b]	Test[c]	Modelled LevelCheck accuracy, %	Source of the frozen value	Published range of the modelled accuracy, %	Smallest passing test count[d]	Test persons; SchwingNet k/n [Wilson]; Holm-adjusted P
H1 (primary)	at risk, pooled	superiority	82.2	a lumbar planning read without a count from C2, coronal localiser [peh1999]: the most favourable published value, measured on all patients	20.8 (count-changing transitional spines counted correctly, 10 of 48 [nakagawa2017]) to 100 (the read counted correctly)	19	**[PENDING G6: H1]**
H2 (key secondary)	at risk, pooled	superiority	97.6	surgeons with the LevelCheck overlay, median of five over 62 difficult films [manbachi2018]; assumes the planning read counted correctly; no interval published	as H1	152	**[PENDING G6: H2]**
H3	at risk, L6	superiority	82.2	as H1 (no subtype given)	17.6 (lumbarised S1 counted correctly, 6 of 34 [nakagawa2017]) to 100	19	**[PENDING G6: H3]**
H4	at risk, sacralised L5	superiority	82.2	as H1	28.6 (sacralisation, Castellvi III, 4 of 14 [nakagawa2017]) to 100	19	**[PENDING G6: H4]**
H5	at risk, T13	superiority	82.2	borrowed from H1: no planning-read series on thoracolumbar variants	31 (a twelve-thoracic count agreeing with morphology on L1, 31 of 100 [yun2018]) to 100	19	**[PENDING G6: H5]**
H6	ordinary spines	non-inferiority, margin 5 points	92.6	97.6 [manbachi2018] minus the margin, chosen knowing the development accuracy on ordinary spines	97.6 to 100 (operating-room series, 20 of 20 [manbachi2018])	48	**[PENDING G6: H6]**
–	transitional, naming unaffected	descriptive	none	97.6 and 100 printed beside	97.6 to 100 (sacralisation, Castellvi II, 23 of 23 counted correctly [nakagawa2017])	–	**[PENDING G6: naming unaffected]**

L5, L6 = fifth, sixth lumbar vertebra; S1 = first sacral segment; T13 = thirteenth thoracic vertebra; LevelCheck = labelling by registration of the preoperative CT to the intraoperative radiograph [levelcheck2012]. [a]Model: LevelCheck carries the labels a reader assigned to the preoperative CT, so its accuracy on a stratum is $r(1-e)$, with r the registration success, frozen at 1.0 (published 0.993–1.000 [levelcheck2016]), and e the probability that the planning read names a scored vertebra otherwise than the reference; no residual surgeon error is multiplied in, both simplifications favouring LevelCheck. Frozen by commit on 23 September 2026 before any held-out transitional person was scored; seen before the freeze were the ordinary-spine development results, the three C-ArmLSpine L6 patients and render L6 recall of 56.1–72.4 %, all below 82.2 % where transitional. [b]Membership from the reference standard alone, fixed before unblinding. At risk: a transitional person with a frame on which a reference-named body is misnamed by counting up from the sacrum over five lumbar vertebrae or down from C2 over seven cervical, twelve thoracic and five lumbar. Naming unaffected: Castellvi I/II, lumbar rib, a fused L6 left unlabelled, a sacralised L5 keeping its label, or no flagged frame. The three C-ArmLSpine L6 patients are excluded from the primary analysis in advance and reported descriptively. [c]Unit: the person. One frame with at least four reference-named bodies drawn per person by a fixed seed from the truth alone; correct when every reference-named body receives its reference name (missed, unnamed, abstained or unanchored = wrong). SchwingNet configuration fixed before unblinding: an arm never trained on a test person, the coupled call, detector confidence 0.74, whole frame, seed 0. One-sided exact binomial P, the probability of at least k correct of n at the frozen value; Holm over the eligible hypotheses at a familywise one-sided 0.025; Wilson 95 % interval for description. [d]The smallest n at which n of n correct gives $P \leq 0.025$; a hypothesis whose stratum is smaller is reported descriptively and not claimed.

Table 4: Level naming by morphology, counting from the detected sacrum, and the coupled call, per stratum and field size.

Mechanism	Test set; stratum; grouping	View; field[b]; operating point	Measure	% (k/n)	Clusters (with an error)	95 % percentile	BCa
Morphology (level model)	C-ArmLSpine, agreed truth[a]; case-grouped folds	lateral; whole frame	exact level, full coverage	98.2 (606/617)	141 (3)	Wilson [94.6, 99.4][g]	[95.0, 99.5]
		AP; whole frame	exact level	96.7 (610/631)	142 (6)	Wilson [92.7, 98.5][g]	[92.9, 98.7]
		lateral; three in field	exact level	92.9 (1,079/1,162)	142 (29)	[89.6, 95.6]	[89.1, 95.4]
		lateral; one in field	exact level	72.8 (440/604)	142 (103)	[69.0, 76.2]	[69.0, 76.2]
	the same; day-grouped folds	lateral; whole frame	exact level	92.5 (571/617)	141	[87.7, 96.4]	[87.0, 96.1]
	the same; no C-ArmLSpine frame in training	lateral; whole frame	exact level	96.3 (594/617)	141 (6)	Wilson [92.0, 98.3][g]	[92.3, 98.5]
		AP; whole frame	exact level	37.9 (239/631)	142 (91)	[29.6, 45.9]	[30.1, 46.4]
	CSXA selection split[f]	cervical lateral	exact level	99.8 (3,629/3,635)	727 patients (3)	Wilson [99.5, 99.9][g]	[99.5, 99.9]
	BUU-LSPINE selection split[f]	lumbar lateral	exact level	97.3 (2,671/2,744)	549 (51)	[96.5, 98.0]	[96.4, 98.0]
	NHANES II selection split[f]	lateral	exact level	99.1 (441/445)	52 patients (4)	Wilson [97.8, 99.6][g]	[98.0, 99.8]
Count from the detected sacrum, no classifier[c]	C-ArmLSpine test, reported detector	whole frame; conf 0.74	counting defined (a sacrum detected)	82.1 (46/56)	28	[69.6, 92.9]	[69.6, 91.1]
			ordinal correct, matched bodies	95.4 (188/197)	28	[89.4, 99.5]	[85.2, 98.9]
			truth bodies at the right ordinal, defined frames	81.0 (188/232)	28	[73.8, 87.6]	[73.5, 87.1]
		whole frame; conf 0.25	ordinal correct, matched bodies	84.7 (199/235)	28	[77.4, 90.7]	[76.5, 90.4]
	COBRA2026 test, held-out detector	conf 0.25	ordinal correct, matched bodies	71.1 (54/76)	27	[56.6, 80.3]	[57.7, 80.6]
	pooled over the three held-out devices	conf 0.25	ordinal correct, matched bodies	81.0 (306/378)	56	[75.7, 85.8]	[75.7, 85.8]
Coupled: classifier and count[d]	C-ArmLSpine folds, agreed truth	whole frame; conf 0.74	classifier alone, counted bodies	98.4 (993/1,009)	130	[96.8, 99.7]	–
			count alone	97.0 (979/1,009)	130	[94.2, 99.2]	–
			coupled	98.6 (995/1,009)	130	[97.2, 99.7]	–
			disagreement frames (the stop signal)	5.2 (12/233)	132	[2.5, 8.7]	–
		whole frame; conf 0.25	classifier alone, counted bodies	98.3 (1,129/1,149)	139	[96.5, 99.6]	–
			count alone	96.1 (1,104/1,149)	139	[93.5, 98.3]	–
			coupled	97.5 (1,120/1,149)	139	[95.7, 99.0]	–
			disagreement frames (the stop signal)	6.8 (18/266)	142	[3.5, 10.2]	–
Transitional stratum	held-out transitional persons of the built split	one frame per person; conf 0.74	all bodies named right (Tables **??**, **??**)	**[PENDING G6: per stratum]**	–	–	–
	three C-ArmLSpine cases read as L6 by one labeller[e]	whole frame; conf 0.25	agrees with that read on the L6 body: coupled; classifier	1 of 5; 0 of 5	3	no interval	–

AP = anteroposterior; BCa = bias-corrected and accelerated bootstrap interval; conf = detector confidence threshold. Intervals are 95 % bootstraps of 2,000 resamples clustered by case (intraoperative) or patient (film); "–" = not computed. [a]Agreed truth = the levels both automatic CBCT labellers name (814 of 910 vertebrae; T11 and most of T12 withheld); ordinary spines. Case-grouped = five folds by case, day-grouped = three folds by acquisition day (a case is an acquisition session, not a proven person); "with an error" = clusters carrying at least one error. [b]Field: whole frame, or crops holding three or one vertebra. Exact level scores every vertebra the film-trained keypoint detector boxed (confidence 0.35; 617 of 676 agreed lateral; with the 59 undetected counted wrong, 87.9 % zero-shot, 89.6 % case-grouped) on its decoded name at full coverage; abstention rows in the Markdown twin. [c]Ordinal = rank of the detected body away from the detected sacrum, scored name-free against the rank from the truth sacrum; defined when truth has a sacrum and one was detected; one-to-one matching at intersection over union ≥ 0.5, whole-vertebra convention (supplement, Table S2); a missed truth body counts as wrong in the defined-frames row. C-ArmLSpine test = 28 cases, 56 AP and lateral frames, device never seen; COBRA2026 test = 27 patients, 609 frames with a sacrum, detector trained on DeepFluoro and C-ArmLSpine real frames; pooled = each real-frame detector on its unseen device. [d]C-ArmLSpine five-fold CV frames with agreed truth (145 cases, 290 frames), the sacrum from the held-out detector. Classifier = the level model's name; count = the ordinary alphabet at the ordinal (L5 at one); coupled = the alphabet (ordinary, L6, T13, both) that best fits the classifier's posteriors at the ordinals; percentile intervals only. [e]Agreement with that read, not correctness: the two labellers segment the same five bodies and disagree only on the names (L1–L5 against L2–L6) on a lumbar-only CBCT; six frames, three clusters, no interval. [f]Patient-keyed validation split, which also chose the encoder checkpoint (one seed); random windows of two to nine bodies (at three: BUU-LSPINE 95.0 %, CSXA 99.6 %, NHANES II 94.6 %); NHANES II 76 films of 52 patients. [g]Wilson 95 % interval on the design-effect-adjusted count where the result file flags errors in few clusters.